Umashankar K H

Forward Deployed Engineer (FDE) Consultant · AI Solutions Architect · Azure & GCP

Bangalore, India · umashankerkh@gmail.com · +91 94836 32823 · umashankar.space · GitHub · linkedin.com/in/umashankar-kh

Professional Profile

Forward Deployed Engineer who sits with the client, understands the business problem, and ships the AI solution to production on **Azure and GCP**, owning the work from scoping and system design through deployment, observability, and cost control. I have delivered multi-agent platforms, RAG systems, voice agents, assessment engines, and accessibility tools across IT services, education, legal, customer support, and enterprise analytics. I also train enterprise engineering teams to do this work, including Claude and Claude Code programmes and a Context Engineering and RAG programme for corporate data and engineering teams, so I explain trade-offs in plain language and leave client teams able to run what I build. M.Tech in Data Science & Machine Learning (PES University).

What I Bring to a Client Engagement

- **Azure and GCP solution provider:** I design and deploy on Azure (Azure OpenAI / AI Foundry, AI Search, Cosmos DB, Container Apps, AKS, Static Web Apps, Logic Apps, Entra ID, Azure ML, Azure DevOps) and GCP (Vertex AI, Gemini, Cloud Run), and I pick the platform that fits the client's stack, compliance needs, and budget.
- **System design:** I architect the full path from ingestion, retrieval, and agent orchestration to guardrails, human sign-off, identity (OAuth2/OIDC, JWT), observability, and CI/CD, with the trade-offs between latency, accuracy, security, and scale written down for the client.
- **Cost economics:** I treat cost as a design input. I work through token spend, KV caching, vector-search cost, cross-provider fallback, and local open-source models as a no-API-cost option, and I built a cost and observability dashboard for FDEs.
- **Multi-sector delivery:** I have deployed AI solutions for IT support and service desks, legal, customer support, accessibility, agent-to-agent (A2A) protocol systems, and data engineering and data science teams.

- **Production discipline:** Evaluation gates (RAGAS, DeepEval), tracing (LangSmith, Langfuse), drift monitoring (Evidently), guardrails for PII and hallucination, and blue-green releases, so the system keeps working after I leave.

Forward Deployed Engineering Experience

FDE Consultant, Chennaya Technologies Pvt Ltd · 2026 to present

- Deploy AI agents and automation for clients, working directly with their teams on scoping, architecture, rollout, and handover.
- Built the **Chennaya AI Observability & Cost Engine**, a dashboard where FDEs analyse model cost frameworks (KV caching and other optimisations) from uploaded run logs, with Gemini-powered analysis and live Azure Log Analytics feeds (Azure OpenAI, App Insights, AKS, errors).

AI Lead Engineer (freelance), MMSMETA · Nov 2025 to May 2026

- Led the project end to end, delivering a RAG platform and AI-driven process automation for internal products on Azure, GCP, and open-source components.

Selected Solutions Delivered

Solution	Sector	Cloud and stack	What it does
Enterprise Support & Ticket Resolution Agent	IT services, enterprise support	Azure OpenAI, Azure AI Search, Cosmos DB, Container Apps, Logic Apps, Entra ID	8-agent pipeline (intake, triage, RAG retrieval, resolution, review, orchestration, action, escalation). Answers are grounded in the knowledge base, guardrailed, and signed off by a human. Live email channel through Logic Apps and Outlook, JWT-secured APIs, and a live admin dashboard with usage analytics.
Shivas, voice-first L1 support agent (M.Tech dissertation)	Customer and technical support	Gemini API, React, TypeScript, Firebase	Under 200 ms first-token latency, 95% intent-categorisation accuracy, and 92%+ first-contact resolution, with automatic silence detection.

Solution	Sector	Cloud and stack	What it does
Inbound call automation and VM provisioning chatbot	Support operations, cloud operations	Python, NLP, Speech APIs, Azure APIs	AI first-level support with intent recognition and probing questions. The chatbot creates, configures, and hands over VMs through conversation, cutting provisioning from hours to minutes.
Enterprise RAG platform (reference build)	Cross-industry knowledge management	GCP: Vertex AI, Gemini Pro, Pinecone, Cloud Run, GitHub Actions	Ingestion, embeddings, retrieval with reranking, and generation, with RAGAS evaluation gates in CI/CD, LangSmith tracing, and PII and hallucination guardrails.
Fine-tuned domain assistant on Azure (reference build)	Enterprise AI platform	Azure ML, Azure OpenAI, AKS, Azure DevOps, MLflow	LoRA fine-tuning pipeline, blue-green deployment on AKS, continuous evaluation with DeepEval, and drift-triggered retraining with Evidently.
Multi-agent workflow automation	Business process automation	LangGraph, n8n, Claude API, Gemini, Redis, Cloud Run, Azure Container Apps	Planner, researcher, coder, and reviewer agents with shared state and cross-provider fallback, traced in Langfuse.

Solution	Sector	Cloud and stack	What it does
Blind-assistance campus navigator	Accessibility, higher education	Android, voice interface	Voice navigation for blind students across an 11-floor, 10-acre campus using floor plans and a 3D model. I lead a team of learners on the build.
Smart Assignment Evaluator	Higher education (Bangalore University)	Gemini API, Google AI Studio, structured output	Rubric-driven grading with JSON-schema outputs, calibrated across subjects, which cuts faculty grading load.
AI Lawyer	Legal	Gemini API, LangChain, vector DB, Firebase	RAG assistant with legal-aware chunking, hybrid retrieval, grounded answers to limit hallucination, and jurisdiction-aware prompts.
Student support chatbots (Innomatics, Eduleem)	EdTech	Azure OpenAI (Foundry), open-source stack, Vercel	Login-gated student support assistants. The Eduleem build uses open-source tools only, to keep running cost low.
Local GenAI assistant	Privacy-sensitive and cost-sensitive teams	Ollama, Llama 3, LangChain, ChromaDB, FastAPI, Docker	Fully offline RAG assistant with zero API cost, shown as an alternative to paid cloud LLMs.

Solution	Sector	Cloud and stack	What it does
Agent-calling system architecture (A2A and MCP)	Data engineering, data science, enterprise automation	A2A protocol, MCP, LangGraph, Claude API, Gemini, Redis, Azure Container Apps, Cloud Run	System architecture in which specialist agents publish their capabilities, discover one another, and call each other over A2A, while MCP connects each agent to its tools and data. A central orchestrator handles routing and shared state, with cross-provider fallback, human sign-off on risky actions, and per-call tracing.
Chennaya AI Observability and Cost Engine (inference cost technology)	AI platform teams, FDEs	Gemini, Azure Log Analytics, App Insights, Google Cloud Run	Takes model run logs, uploaded or live from Azure OpenAI, AKS, and App Insights, and breaks inference cost down per request by token type. It models the savings from KV caching and other optimisations so a team can compare options before choosing one, and it turns the findings into a cost plan for the client.

FDE Enablement and Technical Training

About 5 years of corporate and academic training. Because I teach FDE practice, I stay current on how these systems fail in production.

- **FDE Academy and Futurense (IIT Roorkee, IIT GN-X):** FDE masterclasses and onsite MLOps and LLMOps training on Azure and GCP, including Azure ML pipelines, AKS, Vertex AI RAG, LangGraph agents, and Cloud Run. I run hands-on Azure and GCP labs (static and web apps, storage, databases, connecting live data to an LLM) and the lateral-entry track for engineers with 10 to 15 years of experience, built around an enterprise support-desk reference build.
- **Tiger Analytics (via learning partner):** A phased Claude and Claude Code programme for 20 experienced data engineers, data scientists, and full-stack developers, with track-specific sessions (EDA and feature ideation for data science, schema documentation and data-quality rules for data engineering, code review, security, and test generation for full-stack) and agent, MCP, and skills capstones. I designed it to stay within the client's existing subscription, so cost optimisation was part of the programme.
- **Context Engineering and RAG, corporate programme (Level 2):** Modules on embeddings and FAISS, document ingestion and chunking, context taxonomy and context specs, RAG evaluation with RAGAS, Neo4j knowledge graphs and when to choose graph RAG over vector RAG, and advanced retrieval (HyDE, query rewriting, MMR, hybrid retrieval, re-ranking). Each module had a hands-on lab, ending in an enterprise RAG search engine capstone.
- **Innomatics Research Labs, Eduleem, Great Learning (PES M.Tech), Simplilearn, CAIAS Christ Academy:** Curricula from Python and SQL through ML, NLP, GenAI, agentic AI, and cloud deployment; Microsoft AI-900, AI-102, and AI-104 certification preparation; and Gen AI and LLM faculty sessions.
- **IIRIS Consulting and GAFA:** Faculty for the Certified AI Investigator course, a 48-hour programme for fraud examiners and forensic accountants.
- **Faculty and career sessions:** AI for Bangalore University deans and heads of department across all streams, and speaker at career-guidance events for final-year BCA and MCA students.

Technical Skills

Area	Tools and platforms
Azure	Azure OpenAI and AI Foundry, AI Search, Cosmos DB, Container Apps, AKS, Static Web Apps, Logic Apps, Entra ID, Azure ML, Azure DevOps, Log Analytics, App Insights
GCP	Vertex AI, Gemini, Cloud Run
GenAI and agents	Claude API and Claude Code, MCP, RAG (LangChain, LlamaIndex, pgvector, Pinecone, ChromaDB, FAISS), context engineering, Neo4j graph RAG, HyDE, hybrid retrieval and re-ranking, LangGraph, AutoGen, CrewAI, n8n, LoRA/PEFT
MLOps and LLMOps	Docker, GitHub Actions, MLflow, FastAPI, Evidently, RAGAS, DeepEval, LangSmith, Langfuse, Prometheus, Grafana
System design and cost	Reference architectures, guardrails (NeMo), identity and access, TCO analysis, token and KV-cache cost modelling, local open-source LLMs (Ollama)
Data and languages	Python, SQL, Power BI, Tableau, PySpark, TensorFlow, Airflow

Earlier Experience

- **Lead Learning Designer, Simplilearn** (Oct 2024 to Jan 2025): Designed AI/ML programmes using ADDIE and led Agile delivery with subject-matter experts.
- **Analyst III, Infrastructure Services, DXC Technology** (May 2023 to Oct 2024): Single point of contact for an SAP client's technical issues and owner of the ticket lifecycle. This is the source of my service-desk automation case studies.
- **Quality Analyst, CRM and Lead Systems, Intellipaat** (Aug 2021 to Dec 2022): Built automated lead pipelines in Zoho CRM, Tableau dashboards, and data-quality audits.

Education

- **M.Tech, Data Science and Machine Learning**, PES University, 2026 (CGPA 8.06)